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Section I

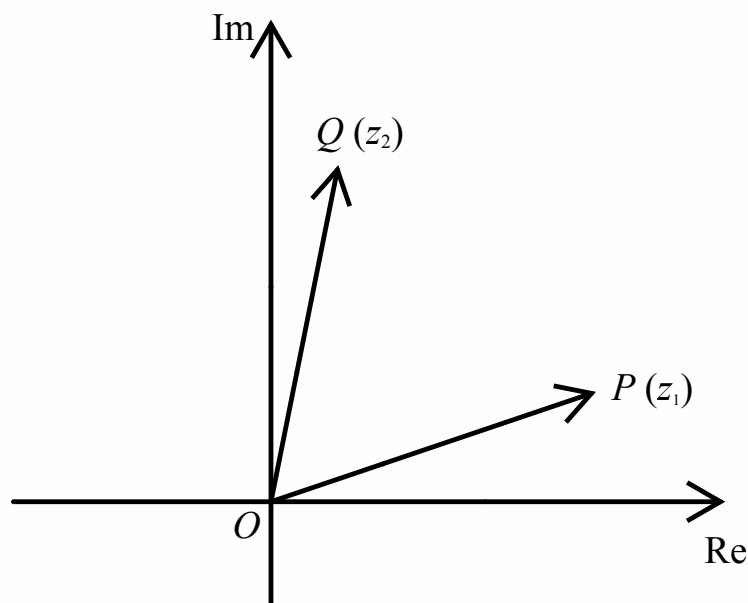
19 marks

Attempt Questions 1–13

Multiple Choice Use the multiple-choice answer sheet for Questions 1–10

- 1 If $z = 3 + 4i$, what is the value of $z\bar{z}$?
- (A) -5 (B) 5 (C) 25 (D) $-7 + 24i$
- 2 What is the value of $(2\cos 110^\circ + 2i \sin 110^\circ)(4\cos 65^\circ + 4i \sin 65^\circ)$?
- (A) $6\cos 175^\circ + 6i \sin 175^\circ$ (B) $8\cos 175^\circ + 8i \sin 175^\circ$
(C) $6\cos 45^\circ + 6i \sin 45^\circ$ (D) $8\cos 45^\circ + 8i \sin 45^\circ$
- 3 What is $-1 - \sqrt{3}i$ expressed in modulus-argument form?
- (A) $\sqrt{2}\text{cis}\left(-\frac{\pi}{3}\right)$ (B) $2\text{cis}\left(-\frac{\pi}{3}\right)$
(C) $\sqrt{2}\text{cis}\left(-\frac{2\pi}{3}\right)$ (D) $2\text{cis}\left(-\frac{2\pi}{3}\right)$
- 4 What is the value of $\left| \frac{(4+2i)(8-6i)}{(3-i)(3+4i)} \right|$?
- (A) 2 (B) $2\sqrt{2}$ (C) 4 (D) $4\sqrt{2}$
- 5 Evaluate i^{-2013}
- (A) 1 (B) i (C) -1 (D) $-i$
- 6 Evaluate $(1 + \sqrt{3}i)^{2014}$
- (A) $2^{2013}(1 + \sqrt{3}i)$ (B) $2^{2013}(1 - \sqrt{3}i)$
(C) $2^{2013}(-1 + \sqrt{3}i)$ (D) $2^{2013}(-1 - \sqrt{3}i)$

- 7 Which of the following graphs is the locus of the point P representing the complex number z in an Argand diagram such that $|z - 2i| = 2 + \text{Im}z$.
- (A) a circle (B) a hyperbola
(C) a parabola (D) a straight line
- 8 If $(a + ib)^2 = 7 + 24i$, where a and b are real numbers, what is the value of $|a| + |b|$?
- (A) 1 (B) 5 (C) 7 (D) 11
- 9 Which expression is a correct factorisation of $z^3 - i$?
- (A) $(z - i)(z^2 + iz + 1)$ (B) $(z + i)(z^2 - iz - 1)$
(C) $(z + i)(z - i)^2$ (D) $(z + i)^3$
- 10 The points P and Q in the first quadrant represent the complex numbers z_1 and z_2 respectively, as shown in the diagram below.
Which statement about the complex number $z_2 - z_1$ is true?

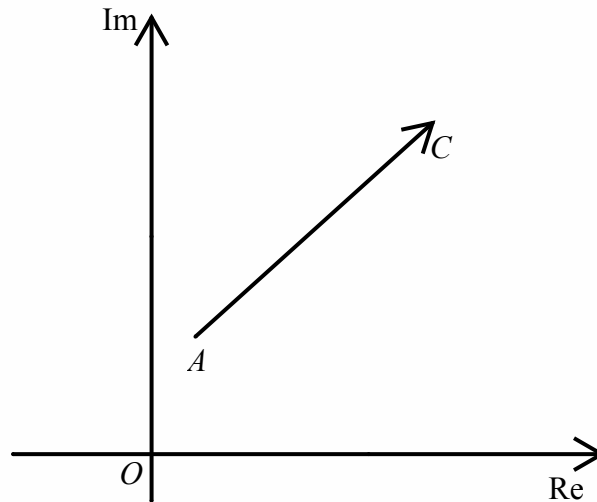


- (A) It is represented by the vector $\overrightarrow{QP}$.
- (B) Its principal argument lies between $\frac{\pi}{2}$ and π .
- (C) Its real part is positive.
- (D) Its modulus is greater than $|z_2 + z_1|$.

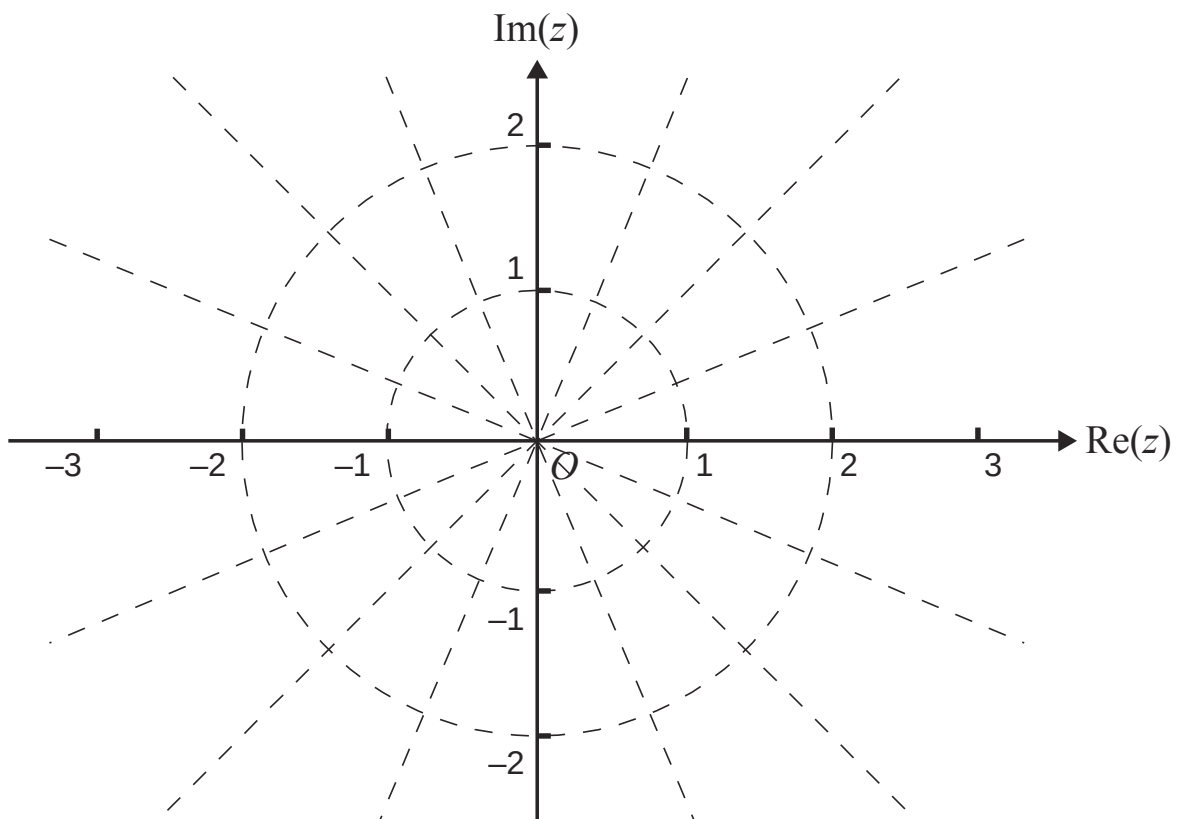
Constrained Answer

For the constrained answers, indicate your answer by using the diagram provided.

- 11 In the diagram below, the vector $\overrightarrow{AC}$ represents the complex number z . 1
 Mark on the diagram a point B , so that the vector $\overrightarrow{AB}$ represents the complex number kz where k is a real number with $0 < k < 1$.



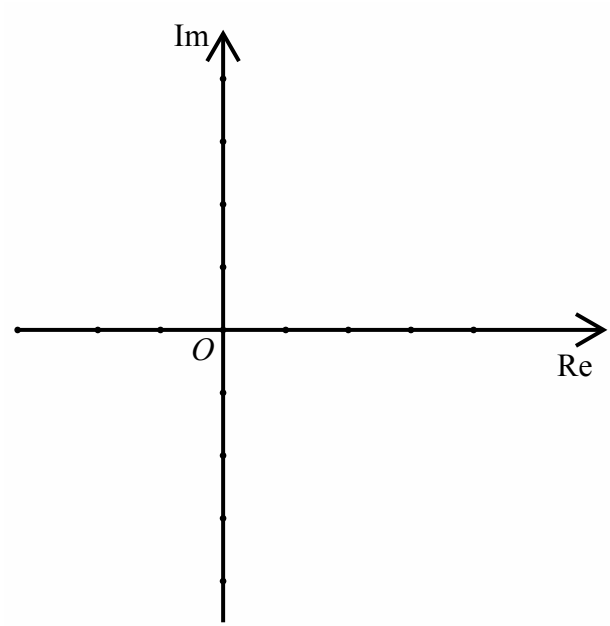
- 12 On the diagram below, mark in the solutions to the equation $z^4 = -16i$ 2



Section I Constrained Response (continued)

- 13 (a) Shade the region in the Argand diagram where $|z + 3i| > 2|z|$

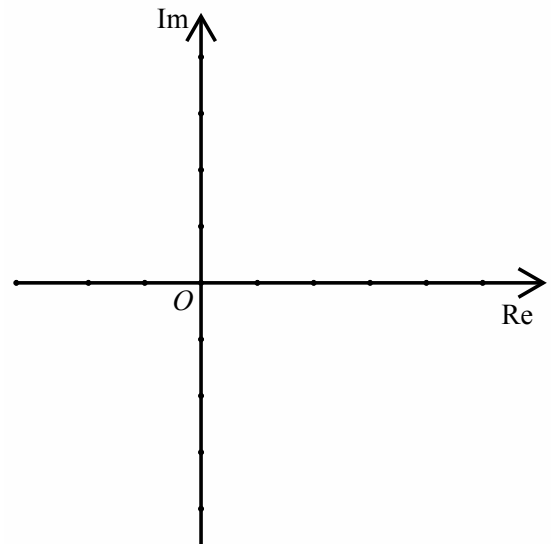
3



- (b) A complex number z satisfies $\arg(z - 1) = \frac{\pi}{6}$.

- (i) Sketch the locus of z .

1



- (ii) Show that $|z - 5| \geq 2$

2

End of Section I

Section II**29 marks****Attempt Questions 14 and 15**

Answer each question in a SEPARATE writing booklet. Extra pages are available.

In Questions 14 and 15 your responses should include relevant mathematical reasoning and/or calculations.

Question 14 (12 Marks) Use a SEPARATE writing booklet.

(a) Let $z = 3 + i$ and $w = 1 - i$. Find in the form $x + iy$:

(i) $2z + iw$. 2

(ii) $\bar{z}w$. 2

(iii) $\frac{6}{w}$. 1

(b) Find the complex number z such that $i(z + 7) + 3(\bar{z} - i) = 0$ 3

(c) Given $z = r(\cos \theta + i \sin \theta)$ where $z \neq 0$.

(i) Show that $z\bar{z}$ is real. 1

(ii) Use de Moivre's theorem to show that $z^n + \bar{z}^n$ is real for all integers $n \geq 1$. 2

(iii) Show that $\frac{z}{\bar{z}} + \frac{\bar{z}}{z}$ is real. 1

Question 15 (17 Marks) Use a SEPARATE writing booklet.

(a) Given $z_1 = i\sqrt{2}$ and $z_2 = \frac{2}{1-i}$

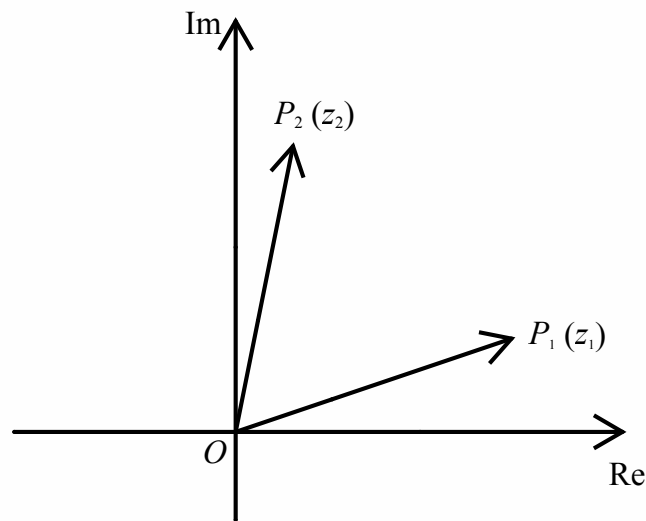
(i) Express z_1 and z_2 in modulus-argument form. 2

(ii) If $z_1 = wz_2$, find the complex number w in modulus-argument form. 1

(iii) On an Argand diagram plot the points P and Q representing the complex numbers z_1 and z_2 respectively. 3
Also show the point R representing $z_1 + z_2$.

(iv) Show that $\arg(z_1 + z_2) = \frac{3\pi}{8}$ and use the diagram to find the exact value of $\tan \frac{3\pi}{8}$. 3

(b) The points P_1 and P_2 represent the complex numbers z_1 and z_2 on the Argand diagram.



(i) Prove that $|z_1 - z_2| \geq |z_1| - |z_2|$. 1

(ii) Hence, or otherwise, if $\left|z - \frac{4}{z}\right| = 2$, prove that the maximum value of $|z|$ is $\sqrt{5} + 1$. 2

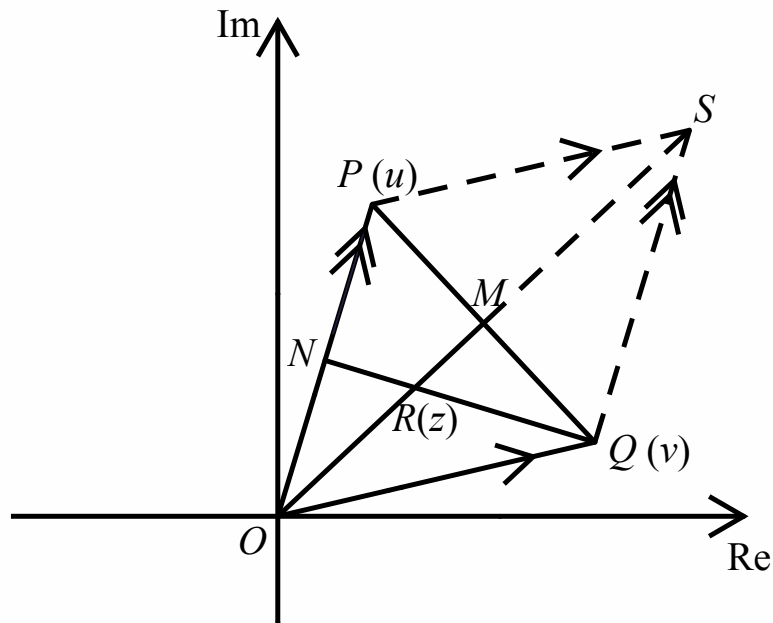
Question 15 continues on page 7

Question 15 (continued)

- (c) In the Argand diagram below, P and Q represent complex numbers u and v respectively. O , P and Q are not collinear.

In $\triangle OPQ$, the line from O to the midpoint M of PQ , meets the line from Q to the midpoint N of OP in the point R . Let R represent the complex number z .

S is the point such that $OPSQ$ is a parallelogram.



- (i) Explain why there are positive real numbers k and l such that

2

$$kz = \frac{1}{2}(u + v) \text{ and } l(z - v) = \frac{1}{2}u - v$$

Hint: Consider question 11

- (ii) Hence show $(3 - 2l)v = 2(k - l)z$.

1

- (iii) Deduce that $z = \frac{1}{3}(u + v)$.

2

End of paper



NORTH SYDNEY GIRLS HIGH SCHOOL

2013

HSC Mathematics Extension 2

Term 4 Extension 2 Assessment Task 1

SOLUTIONS

Section I Multiple Choice

1. ☐ A ☐ B ☒ C ☐ D

2. ☐ A ☒ B ☐ C ☐ D

3. ☐ A ☐ B ☐ C ☒ D

4. ☐ A ☒ B ☐ C ☐ D

5. ☐ A ☐ B ☐ C ☒ D

6. ☐ A ☐ B ☐ C ☒ D

7. ☐ A ☐ B ☒ C ☐ D

8. ☐ A ☐ B ☒ C ☐ D

9. ☐ A ☒ B ☐ C ☐ D

10. ☐ A ☒ B ☐ C ☐ D

Section I Multiple Choice

1 If $z = 3 + 4i$, what is the value of $z\bar{z}$?

- (A) -5 (B) 5 (C) 25 (D) $-7 + 24i$

$$z = 3 + 4i \Rightarrow |z| = 5$$

$$z\bar{z} = |z|^2 = 25$$

2 What is the value of $(2\cos 110^\circ + 2i \sin 110^\circ)(4\cos 65^\circ + 4i \sin 65^\circ)$?

- (A) $6\cos 175^\circ + 6i \sin 175^\circ$ (B) $8\cos 175^\circ + 8i \sin 175^\circ$

- (C) $6\cos 45^\circ + 6i \sin 45^\circ$ (D) $8\cos 45^\circ + 8i \sin 45^\circ$

$$r\text{cis}\alpha \times R\text{cis}\beta = rR\text{cis}(\alpha + \beta)$$

3 What is $-1 - \sqrt{3}i$ expressed in modulus-argument form?

- (A) $\sqrt{2}\text{cis}\left(-\frac{\pi}{3}\right)$ (B) $2\text{cis}\left(-\frac{\pi}{3}\right)$

- (C) $\sqrt{2}\text{cis}\left(-\frac{2\pi}{3}\right)$ (D) $2\text{cis}\left(-\frac{2\pi}{3}\right)$

4 What is the value of $\left| \frac{(4+2i)(8-6i)}{(3-i)(3+4i)} \right|$?

- (A) 2 (B) $2\sqrt{2}$ (C) 4 (D) $4\sqrt{2}$

$$\left| \frac{wz}{u} \right| = \frac{|w| \times |z|}{|u|}$$

$$\therefore \left| \frac{(4+2i)(8-6i)}{(3-i)(3+4i)} \right| = \frac{|4+2i| \times |8-6i|}{|3-i| \times |3+4i|} = \frac{\sqrt{20} \times 10}{\sqrt{10} \times 5} = 2\sqrt{2}$$

5 Evaluate i^{-2013}

- (A) 1 (B) i (C) -1 (D) $-i$

$$i^{2013} = i^{2012} \times i = i$$

$$\therefore i^{-2013} = i^{-1} = -i$$

6 Evaluate $(1 + \sqrt{3}i)^{2014}$

- (A) $2^{2013}(1 + \sqrt{3}i)$ (B) $2^{2013}(1 - \sqrt{3}i)$

- (C) $2^{2013}(-1 + \sqrt{3}i)$ (D) $2^{2013}(-1 - \sqrt{3}i)$

$$(1 + \sqrt{3}i)^{2014} = (2\text{cis}\frac{\pi}{3})^{2014} = 2^{2014}\text{cis}\frac{2014\pi}{3} = 2^{2014}\text{cis}\left(-\frac{2\pi}{3}\right)$$

$$= 2^{2014}\left(-\frac{1}{2} - i\frac{\sqrt{3}}{2}\right) = 2^{2013}(-1 - \sqrt{3}i)$$

- 7 Which of the following graphs is the locus of the point P representing the complex number z in an Argand diagram such that $|z - 2i| = 2 + \text{Im}z$.

(A) a circle (B) a hyperbola
(C) a parabola (D) a straight line

Focus-directrix definition: distance from a fixed point $(0, 2)$ is equal to the distance from the fixed line $(y = -2)$.

- 8 If $(a + ib)^2 = 7 + 24i$, where a and b are real numbers, what is the value of $|a| + |b|$?

(A) 1 (B) 5 (C) 7 (D) 11

Expanding the LHS and equating real and imaginary parts yields: $a^2 - b^2 = 7, ab = 12$
By inspection $a = 4, b = 3$.

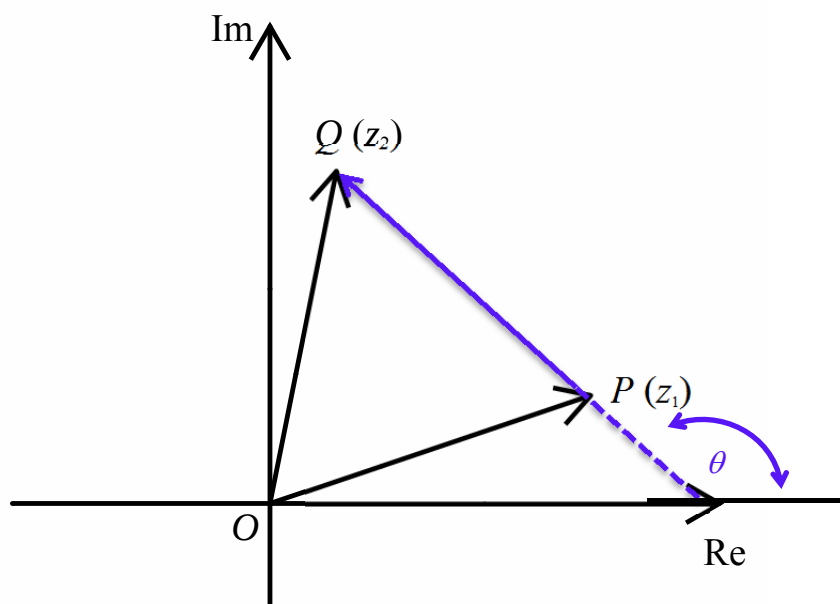
- 9 Which expression is a correct factorisation of $z^3 - i$?

(A) $(z - i)(z^2 + iz + 1)$ (B) $(z + i)(z^2 - iz - 1)$

(C) $(z + i)(z - i)^2$ (D) $(z + i)^3$
 $z^3 - i = z^3 + i^3 = (z + i)(z^2 - iz + i^2) = (z + i)(z^2 - iz - 1)$

- 10 The points P and Q in the first quadrant represent the complex numbers z_1 and z_2 respectively, as shown in the diagram below.

Which statement about the complex number $z_2 - z_1$ is true?



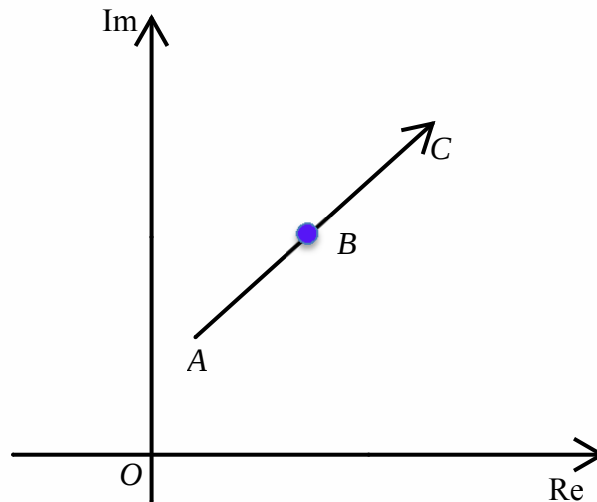
- (A) It is represented by the vector $\overrightarrow{QP}$.
(B) Its principal argument lies between $\frac{\pi}{2}$ and π .
(C) Its real part is positive.
(D) Its modulus is greater than $|z_2 + z_1|$.

From the diagram above, $\arg(z_2 - z_1) = \theta$ and $\frac{\pi}{2} < \theta < \pi$.

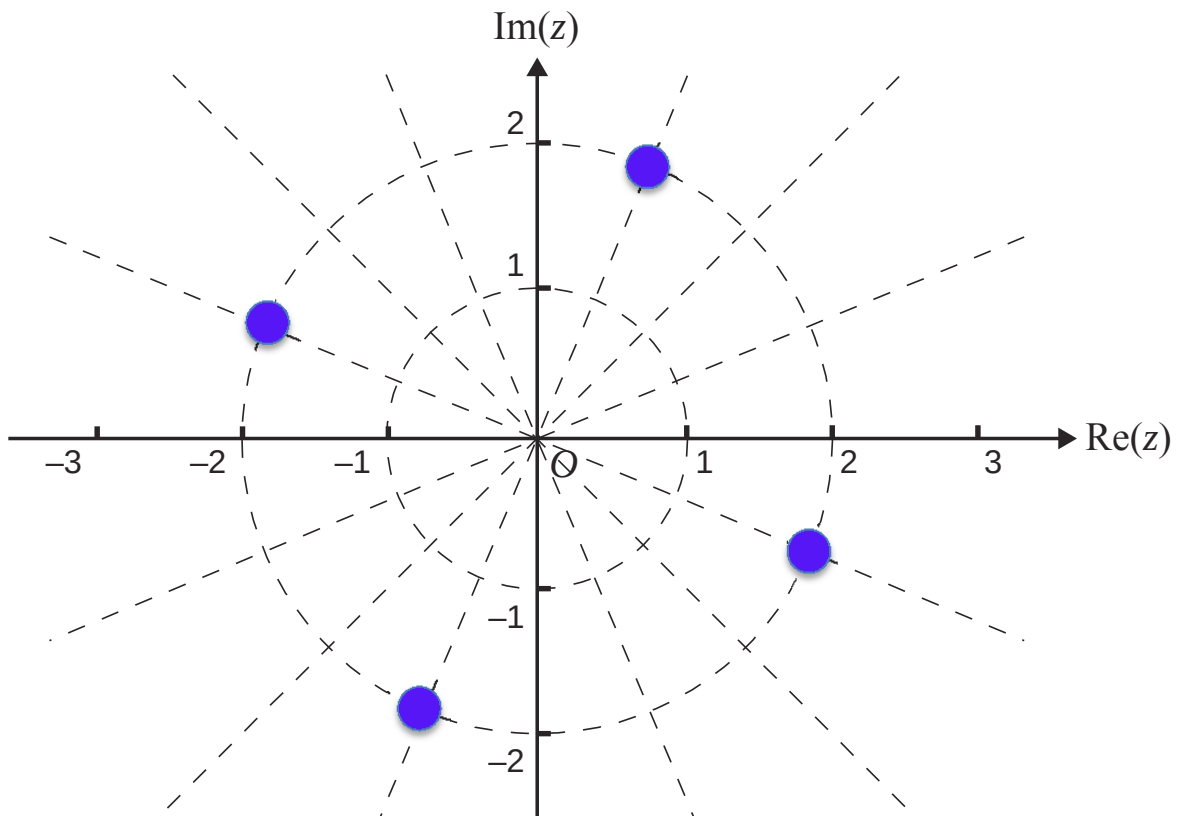
Constrained Answer

For the constrained answers, indicate your answer by using the diagram provided.

- 11 In the diagram below, the vector $\overrightarrow{AC}$ represents the complex number z . 1
 Mark on the diagram a point B , so that the vector $\overrightarrow{AB}$ represents the complex number kz where k is a real number with $0 < k < 1$.



- 12 On the diagram below, mark in the solutions to the equation $z^4 = -16i$ 2
 As $z^4 = 16\text{cis}(-\frac{\pi}{2})$ then a solution would be $z = 2\text{cis}(-\frac{\pi}{8})$. The rest are equally spaced.



Section I Constrained Response (continued)

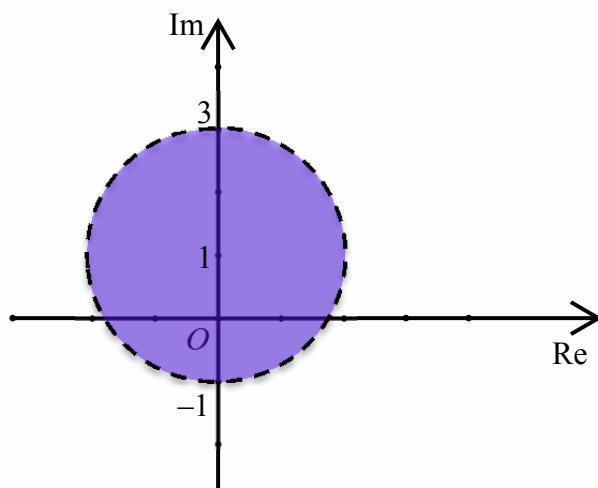
- 13 (a) Shade the region in the Argand diagram where $|z + 3i| > 2|z|$ 3

$$|z + 3i| = \sqrt{x^2 + (y + 3)^2}$$

$$2|z| = 2\sqrt{x^2 + y^2}$$

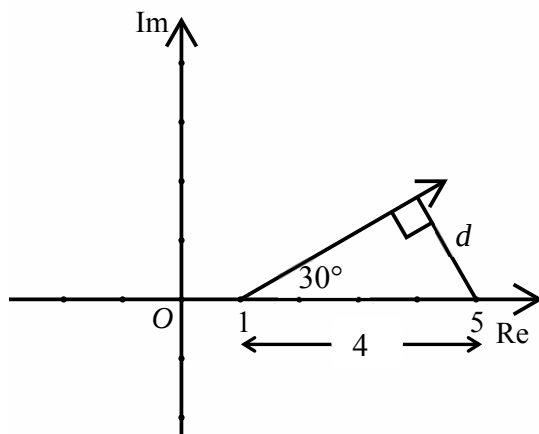
$$\therefore x^2 + (y + 3)^2 > 4(x^2 + y^2) \Rightarrow 3x^2 + 3y^2 - 6y < 9$$

$$\therefore x^2 + y^2 - 2y < 3 \Rightarrow x^2 + (y - 1)^2 < 4$$



- (b) A complex number z satisfies $\arg(z - 1) = \frac{\pi}{6}$.

- (i) Sketch the locus of z . 1



- (ii) Show that $|z - 5| \geq 2$ 2

The minimum distance from $z = 5$ to the ray is the perpendicular distance, d .

$$\therefore \sin 30^\circ = \frac{d}{4}$$

$$\therefore d = 4 \sin 30^\circ$$

$$\therefore d = 2$$

$$\therefore |z - 5| \geq 2$$

End of Section I

Section II

Question 14 (12 Marks)

(a) Let $z = 3 + i$ and $w = 1 - i$. Find in the form $x + iy$:

(i) $2z + iw$. 2

$$\begin{aligned}2z + iw &= 2(3 + i) + i(1 - i) \\&= 6 + 2i + i + 1 \\&= 7 + 3i\end{aligned}$$

(ii) $\bar{z}w$. 2

$$\begin{aligned}\bar{z}w &= (3 - i)(1 - i) \\&= 3 - 1 - 3i - i \\&= 2 - 4i\end{aligned}$$

(iii) $\frac{6}{w}$. 1

$$\begin{aligned}\frac{6}{w} &= \frac{6}{1 - i} \\&= \frac{6(1 + i)}{2} \quad \left[\frac{1}{w} = \frac{\bar{w}}{|w|^2} \right] \\&= 3 + 3i\end{aligned}$$

(b) Find the complex number z such that $i(z + 7) + 3(\bar{z} - i) = 0$ 3

Let $z = x + iy$:

$$\therefore i(x + 7 + iy) + 3(x - iy - i) = 0$$

$$\therefore 3x - y + i(x + 4 - 3y) = 0$$

Comparing real and imaginary parts:

$$\therefore \begin{cases} 3x - y = 0 \\ x - 3y = -4 \end{cases}$$

$$\therefore x - 3(3x) = -4$$

$$\therefore x = \frac{1}{2}$$

$$\therefore y = \frac{3}{2}$$

$$\therefore z = \frac{1}{2}(1 + 3i)$$

(c) Given $z = r(\cos \theta + i \sin \theta)$ where $z \neq 0$.

(i) Show that $z\bar{z}$ is real.

1

$$\begin{aligned} z\bar{z} &= |z|^2 \\ &= |r \operatorname{cis} \theta|^2 \\ &= r^2 \in \mathbb{R} \end{aligned}$$

(ii) Use de Moivre's theorem to show that $z^n + \bar{z}^n$ is real for all integers $n \geq 1$.

2

$$\begin{aligned} \bar{z} &= r \operatorname{cis}(-\theta) \\ z^n + \bar{z}^n &= r^n \operatorname{cis}(n\theta) + r^n \operatorname{cis}(-n\theta) \\ &= r^n [\operatorname{cis}(n\theta) + \operatorname{cis}(-n\theta)] \\ &= 2r^n \cos n\theta \quad \left[w + \bar{w} = 2 \operatorname{Re} w \right] \\ &\in \mathbb{R} \end{aligned}$$

(iii) Show that $\frac{z}{\bar{z}} + \frac{\bar{z}}{z}$ is real.

1

$$\begin{aligned} \frac{z}{\bar{z}} + \frac{\bar{z}}{z} &= \frac{z^2 + (\bar{z})^2}{z\bar{z}} \\ &= \frac{2r^2 \cos 2\theta}{r^2} \quad [\text{From (i) and (ii)}] \\ &= 2 \cos 2\theta \\ &\in \mathbb{R} \end{aligned}$$

Question 15 (17 Marks)

(a) Given $z_1 = i\sqrt{2}$ and $z_2 = \frac{2}{1-i}$

- (i) Express z_1 and z_2 in modulus-argument form.

2

$$z_1 = i\sqrt{2} = \sqrt{2}\text{cis}\frac{\pi}{2}$$

Alternatively:

$$\begin{aligned} z_2 &= \frac{2}{\sqrt{2}\text{cis}\left(-\frac{\pi}{4}\right)} \\ &= \sqrt{2}\text{cis}\frac{\pi}{4} \end{aligned}$$

$$\begin{aligned} z_2 &= \frac{2}{1-i} \\ &= \frac{2(1+i)}{2} \\ &= 1+i \\ &= \sqrt{2}\text{cis}\frac{\pi}{4} \end{aligned}$$

- (ii) If $z_1 = wz_2$, find the complex number w in modulus-argument form.

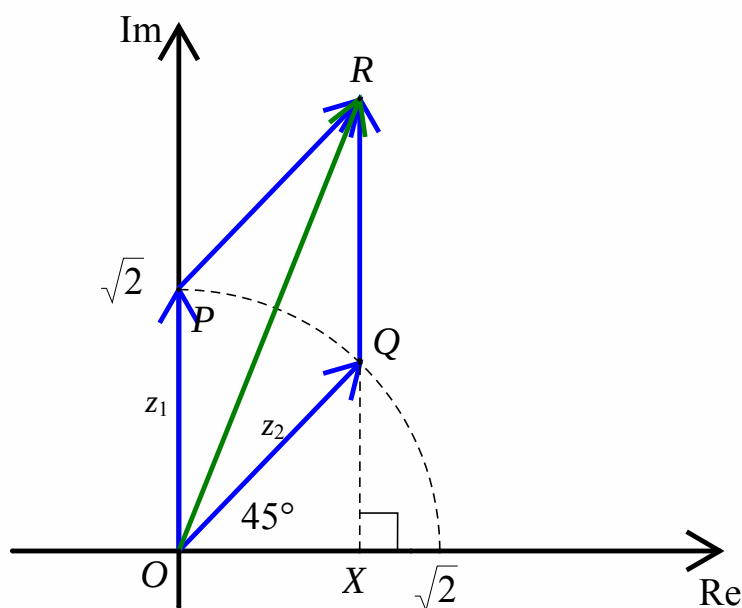
1

$$\begin{aligned} w &= \frac{z_1}{z_2} \\ &= \frac{\sqrt{2}\text{cis}\frac{\pi}{2}}{\sqrt{2}\text{cis}\frac{\pi}{4}} \\ &= \text{cis}\frac{\pi}{4} \end{aligned}$$

- (iii) On an Argand diagram plot the points P and Q representing the complex numbers z_1 and z_2 respectively.

3

Also show the point R representing $z_1 + z_2$.



NB $\overrightarrow{OP} = \overrightarrow{QR}$ and $\overrightarrow{OQ} = \overrightarrow{PR}$

NB Q could be indicated by marking in its coordinates $(1, 1)$.

- (iv) Show that $\arg(z_1 + z_2) = \frac{3\pi}{8}$ and use the diagram to find the exact value of $\tan \frac{3\pi}{8}$.

3

$$\angle QOX = \frac{\pi}{4} \Rightarrow \angle POQ = \frac{\pi}{4}$$

$OPRQ$ is a parallelogram by construction, with $OP = OQ = \sqrt{2}$.
 $\therefore OPRQ$ is a rhombus.

$$\therefore \angle ROQ = \frac{1}{2} \angle POQ = \frac{\pi}{8} \quad (\text{diagonals bisect vertex angles of a rhombus})$$

$$\therefore \angle ROX = \frac{\pi}{4} + \frac{\pi}{8} = \frac{3\pi}{8}$$

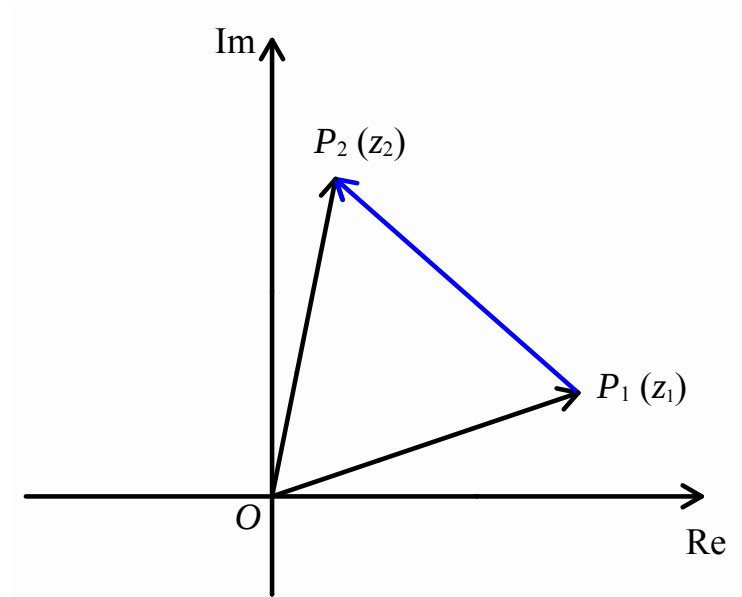
$$\therefore \arg(z_1 + z_2) = \angle ROX = \frac{3\pi}{8}$$

$$\text{From (i), } z_1 + z_2 = i\sqrt{2} + 1 + i = 1 + i(1 + \sqrt{2})$$

$$\therefore \tan[\arg(z_1 + z_2)] = 1 + \sqrt{2}$$

$$\therefore \tan \frac{3\pi}{8} = 1 + \sqrt{2}$$

- (b) The points P_1 and P_2 represent the complex numbers z_1 and z_2 on the Argand diagram.



- (i) Prove that $|z_1 - z_2| \geq |z_1| - |z_2|$.

1

In the diagram above, $OP_2 = |z_2|$, $OP_1 = |z_1|$ and $P_1P_2 = |z_2 - z_1|$

So by the triangle inequality $P_1P_2 + OP_2 > OP_1$.

$$\therefore |z_1 - z_2| + |z_2| > |z_1|$$

$$\therefore |z_1 - z_2| > |z_1| - |z_2|$$

Equality occurs when O , P_1 and P_2 are collinear.

$$\therefore |z_1 - z_2| \geq |z_1| - |z_2|$$

- (ii) Hence, or otherwise, if $\left| z - \frac{4}{z} \right| = 2$, prove that the maximum value of $|z|$ is $\sqrt{5} + 1$.

2

From (i):

$$\begin{aligned} 2 &= \left| z - \frac{4}{z} \right| \\ &\geq \left| z \right| - \left| \frac{4}{z} \right| \\ &= \left| z \right| - \frac{4}{\left| z \right|} \end{aligned}$$

Let $x = |z|$:

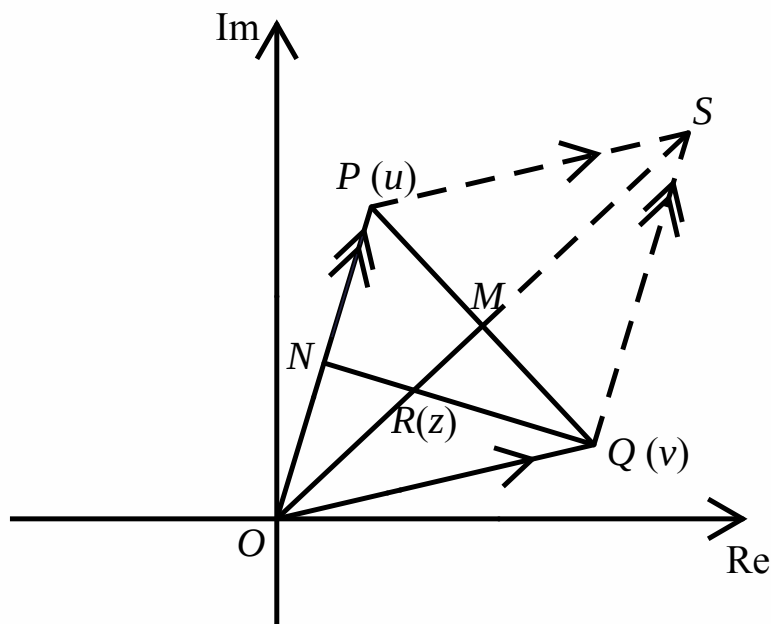
$$\begin{aligned} \therefore 2x &\geq x^2 - 4 \\ \therefore x^2 - 2x - 4 &\leq 0 \\ \therefore (x-1)^2 &\leq 5 \\ \therefore -\sqrt{5} &\leq x-1 \leq \sqrt{5} \\ \therefore x &\leq \sqrt{5} + 1 \end{aligned}$$

$$\therefore |z|_{\max} = \sqrt{5} + 1$$

- (c) In the Argand diagram below, P and Q represent complex numbers u and v respectively. O, P and Q are not collinear.

In $\triangle OPQ$, the line from O to the midpoint M of PQ , meets the line from Q to the midpoint N of OP in the point R . Let R represent the complex number z .

S is the point such that $OPSQ$ is a parallelogram.



- (i) Explain why there are positive real numbers k and l such that

2

$$kz = \frac{1}{2}(u + v) \text{ and } l(z - v) = \frac{1}{2}u - v$$

Hint: Consider question 11

In the diagram above M corresponds to the complex number $\frac{1}{2}(u + v)$.

The vectors OR and OM are coincident i.e. $\arg(\overrightarrow{OR}) = \arg(\overrightarrow{OM})$, so there is a real number k such that $kz = \frac{1}{2}(u + v)$.

Now N corresponds to the complex number $\frac{1}{2}u$ and so the vector QN corresponds to the complex number $\frac{1}{2}u - v$ and vector QR corresponds to $z - v$.

As the vectors QR and QN are coincident i.e. $\arg(\overrightarrow{QR}) = \arg(\overrightarrow{QN})$ there is a real number l such that $l(z - v) = \frac{1}{2}u - v$.

- (ii) Hence show $(3 - 2l)v = 2(k - l)z$.

1

$$kz = \frac{1}{2}u + \frac{1}{2}v \quad (1)$$

$$lz - lv = \frac{1}{2}u - v \quad (2)$$

$$\begin{aligned} (1) - (2): \quad kz - lz + lv &= \frac{3}{2}v \\ \therefore 2kz - 2lz + 2lv &= 3v \\ \therefore 2(k - l)z &= (3 - 2l)v \end{aligned}$$

(iii) Deduce that $z = \frac{1}{3}(u + v)$.

2

From (ii), $(3 - 2l)v = 2(k - l)z$

As $3 - 2l$ and $2(k - l)$ are real numbers this implies that the vectors representing the complex numbers v and z are coincident.

This is impossible as the vectors OR and OQ are not coincident.

$$\therefore 3 - 2l = 2(k - l) = 0$$

$$\therefore k = l = \frac{3}{2}$$

As $kz = \frac{1}{2}(u + v)$ then $\frac{3}{2}z = \frac{1}{2}(u + v)$.

$$\therefore z = \frac{1}{3}(u + v)$$

End of Solutions